MP307 Practical 3 Discrete Population Models

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In [1]:
```
%matplotlib notebook
import numpy as np
import matplotlib.pyplot as plt
```

Q.1(*) Geometric Population Model

Consider the geometric population model $P_{n+1} = (1+r)P_n$ for $n = 0, 1, \ldots$ with solution $P_n = P_0 e^{sn}$ for $s = \log(1+r)$ as a model of

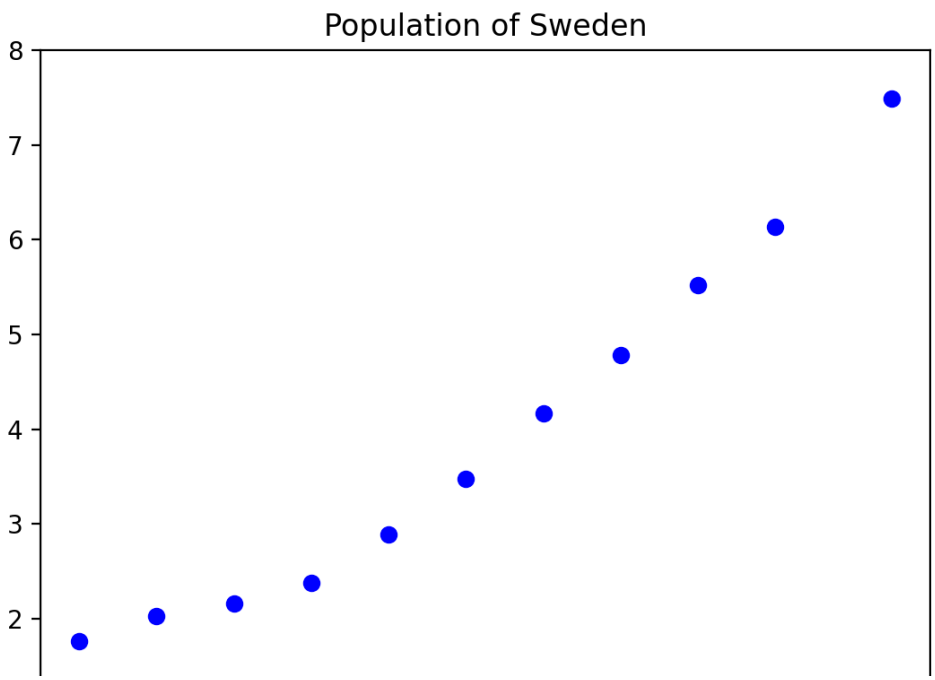
1. Sweden's population with 11 samples from 1750 to 1960
2. The US population with 19 samples from 1790 to 1990.

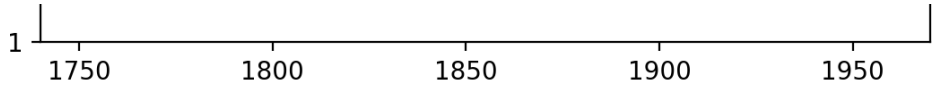
Population of Sweden

The data (in millions) are given in the lists `Year` and `Pop` below.

In [2]:
```
Year = [1750, 1770, 1790, 1810, 1830, 1850, 1870, 1890, 1910, 1930, 1960]
Pop= [1.76, 2.03, 2.158, 2.378, 2.888, 3.48, 4.17, 4.785, 5.522, 6.14, 7.49
```

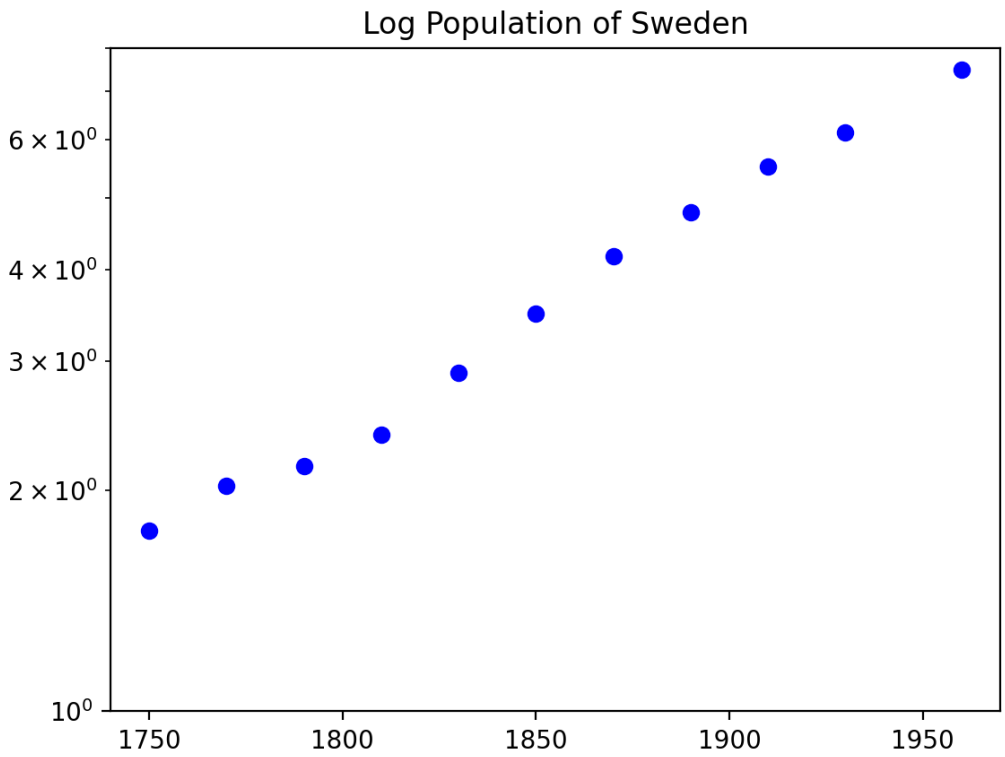
In [3]:
```
plt.figure(1)
plt.plot(Year,Pop,'bo')
plt.axis([1740,1970,1,8])
plt.title('Population of Sweden')
plt.show()
```





Look at t vs log(P):

In [4]:
```
plt.figure(2) # new figure
plt.plot(Year,Pop,'bo')
plt.yscale("log")
plt.axis([1740,1970,1,8])
plt.title('Log Population of Sweden')
plt.show()
```



The log plot suggest a linear relationship to between $\log P$ and t of the form $\log P(t) = st + b$ for some s, b. Thus $P(t) = P_0 e^{st}$ with $P_0 = \exp(b)$.

In [5]:
```
log_Pop=np.log(Pop) # take log of each entry in S_Pop
```

In [6]:
```
print(log_Pop)
```

```
[0.56531381 0.70803579 0.76918187 0.8662598  1.06056422 1.24703229
 1.42791604 1.56548602 1.70874011 1.81482474 2.0135688 ]
```

Use a least squares approximation to find a line as explained in the notes for matching data x_i to y_i.

In [7]:
```
N=len(Year) # no of elements in S_Year
sumx=0; sumy=0; sumxx=0; sumxy=0;
for i in range(N):
    sumx=sumx+Year[i]
    sumy=sumy+log_Pop[i]
    sumxx=sumxx+Year[i]**2
    sumxy=sumxy+log_Pop[i]*Year[i]
Xbar=sumx/N
Ybar=sumy/N
XXbar=sumxx/N
XYbar=sumxy/N
```

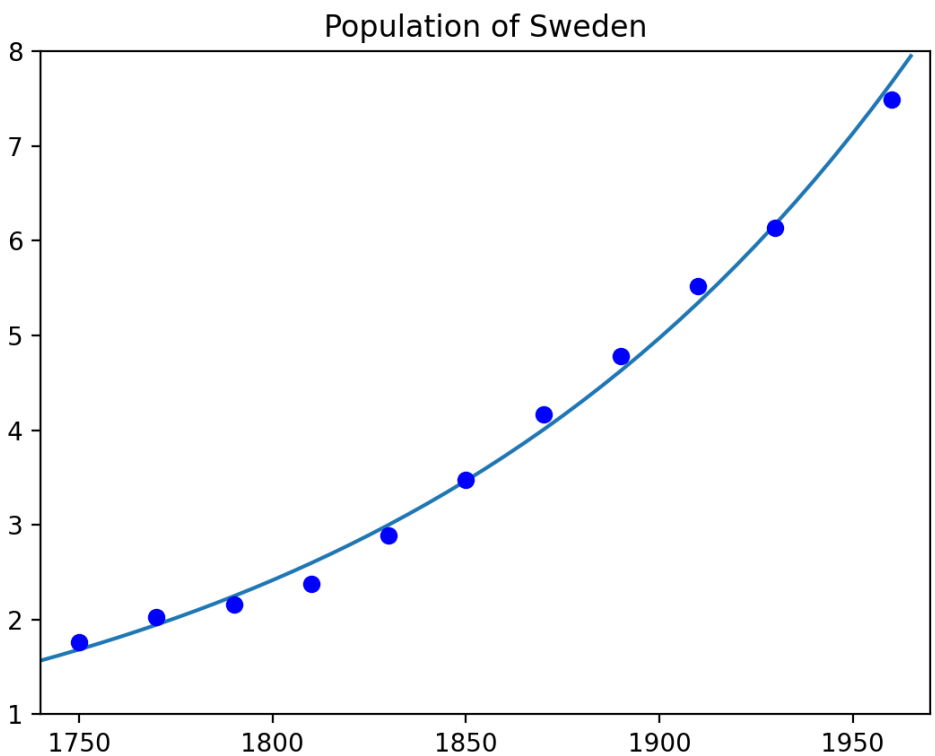
In [8]:
```
s=(XYbar-Xbar*Ybar)/(XXbar-Xbar**2)
b=(XXbar*Ybar-Xbar*XYbar)/(XXbar-Xbar**2)

P_0=np.exp(b)
print(s,P_0)
```

```
0.0072182711579564076 5.500654960458621e-06
```

Plot population $P(t) = P_0 e^{st}$ vs t.

In [9]:
```
t=np.arange(1740.0,1970.,5.)
P_LS=P_0*np.exp(s*t) # least squares approximation
```

In [10]:
```
plt.figure(3)
plt.plot(t,P_LS)
plt.plot(Year,Pop,'bo')
plt.axis([1740,1970,1,8])
plt.title('Population of Sweden')
plt.show()
```



In [ ]:

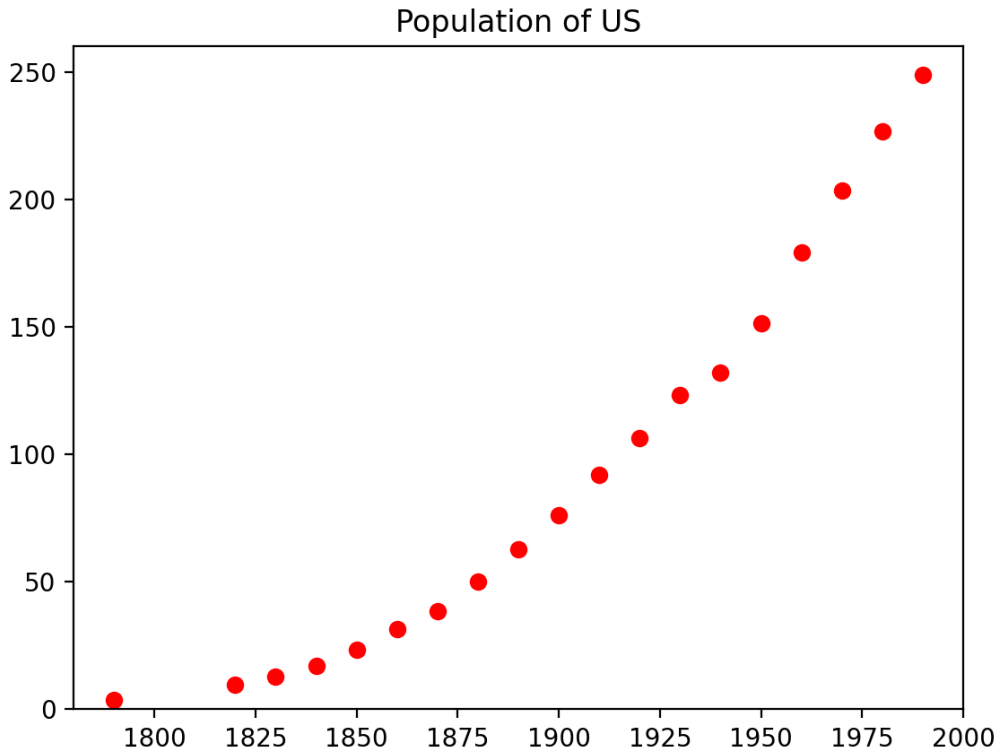
Repeat this analysis for the US population as follows:

In [11]:

```
Year = [1790, 1820, 1830, 1840, 1850, 1860, 1870, 1880, 1890, 1900, 1910, 1
Pop= [3.79, 9.6, 12.9, 17.1, 23.2, 31.4, 38.6, 50.2, 62.9, 76.0, 92.0, 106.
```

In [12]:

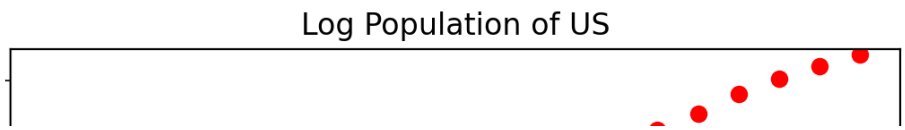
```
plt.figure(4)
plt.plot(Year,Pop,'ro')
plt.axis([1780,2000,0,260])
plt.title('Population of US')
plt.show()
```

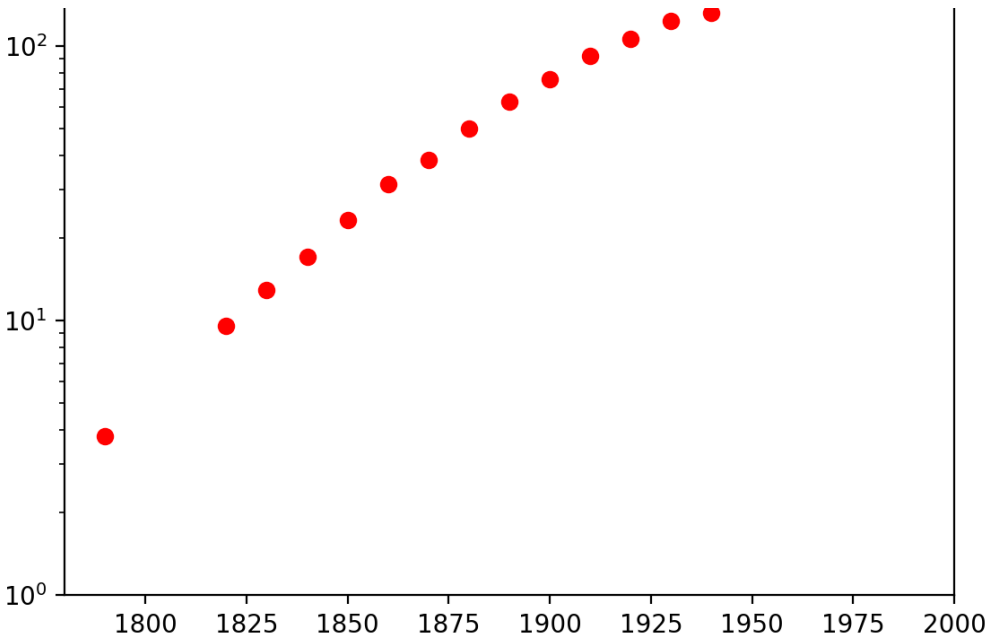


Look at t vs log(P):

In [15]:

```
plt.figure(5) # new figure
plt.plot(Year,Pop,'ro')
plt.yscale("log")
plt.axis([1780,2000,1,260])
plt.title('Log Population of US')
plt.show()
```





In [16]:
```
log_Pop=np.log(Pop)
```

In [17]:
```
print(log_Pop)
```

```
[1.33236602 2.2617631  2.55722731 2.83907846 3.14415228 3.44680789
 3.65325228 3.91601503 4.14154616 4.33073334 4.52178858 4.66814499
 4.81380905 4.88280192 5.01926462 5.18906038 5.31468272 5.42292153
 5.51624735]
```

In [18]:
```
N=len(Year) # no of elements in US_Year
sumx=0; sumy=0; sumxx=0; sumxy=0;
for i in range(N):
    sumx=sumx+Year[i]
    sumy=sumy+log_Pop[i]
    sumxx=sumxx+Year[i]**2
    sumxy=sumxy+log_Pop[i]*Year[i]
Xbar=sumx/N
Ybar=sumy/N
XXbar=sumxx/N
XYbar=sumxy/N
```

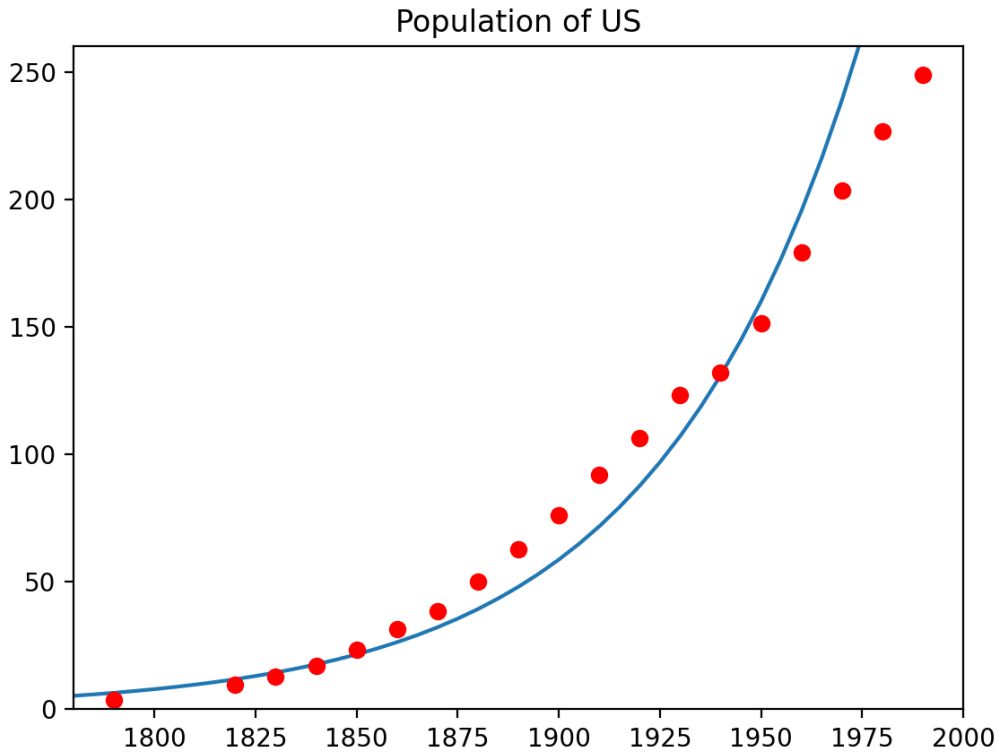
In [20]:
```
s=(XYbar-Xbar*Ybar)/(XXbar-Xbar**2)
b=(XXbar*Ybar-Xbar*XYbar)/(XXbar-Xbar**2)

P_0=np.exp(b)
print(s,P_0)
```

```
0.020051106626461583 1.6718326017274779e-15
```

In [21]:
```
t=np.arange(1780.0,2000.0,5.)
P_LS=P_0*np.exp(s*t) # least squares approximation
```

In [22]:

```
plt.figure(6)
plt.plot(t,P_LS)
plt.plot(Year,Pop,'ro')
plt.axis([1780,2000,0,260])
plt.title('Population of US')
plt.show()
```



Q.2 (*) Discrete Verhulst Model

The general discrete Verhulst model of population growth is as follows

$$P_{n+1} = \frac{1+r}{1+\frac{rP_n}{K}} P_n$$

where $P_n \to K$ as $n \to \infty$.

For the choice of parameters $P_0 = 3.79$, $r = 0.35$ and $K = 300$ compare the Verhulst model to the US population from 1790 to 1990. Explain what you see.

In [23]:

```
print(Year)
```

```
[1790, 1820, 1830, 1840, 1850, 1860, 1870, 1880, 1890, 1900, 1910, 1920, 19
30, 1940, 1950, 1960, 1970, 1980, 1990]
```

In [24]:

```
P_0=3.79
r=0.35
K=300.0
```

In [25]:

```
# plot Verhulst pop PV from 1790 to 1990 with 10 year intervals
PV=[0]*20 # list for Verhulst P initialized with 20 zeros
YV=[0]*20 # list for years initialized with 20 zeros

PV[0]=P_0
YV[0]=1790
# choose 10 year intervals
for n in range(1,20):
    YV[n]=1790+10*n
    PV[n]=(1+r)*PV[n-1]/(1+r*PV[n-1]/K)
```

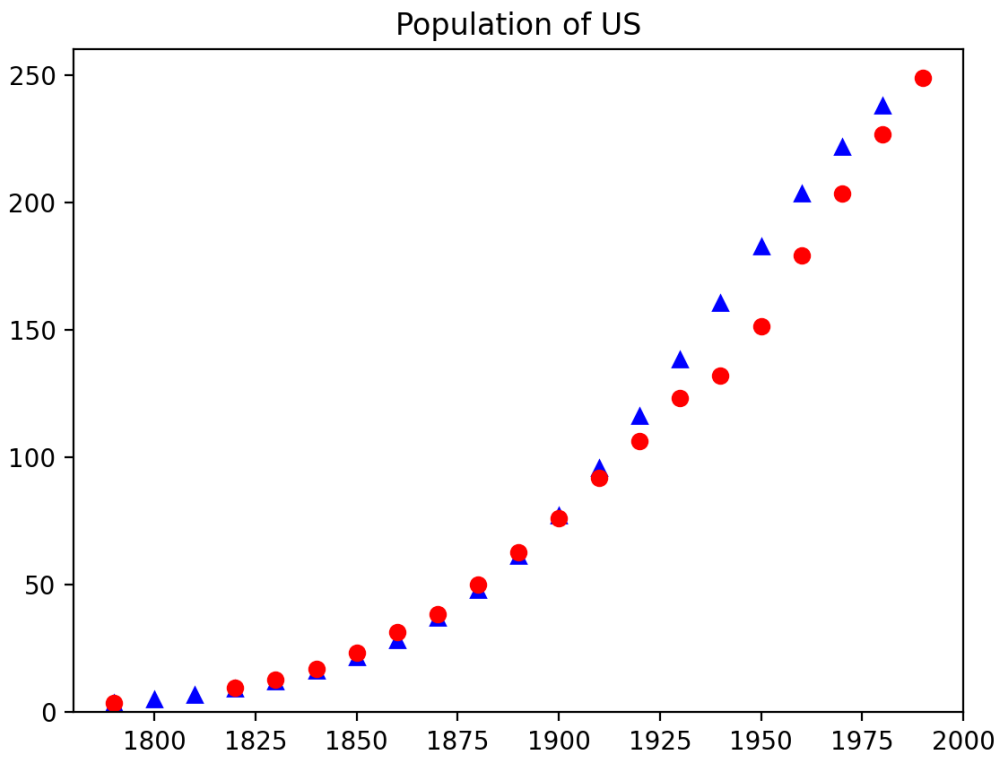
In [26]:

```
print(PV)
```

```
[3.79, 5.093976135520772, 6.836240197791433, 9.15590034404303, 12.229827865
107413, 16.278010802773355, 21.56575898309768, 28.39924806493404, 37.109457
41970735, 48.0188229724253, 61.386423256309854, 77.33325731948169, 95.76020
718123476, 116.28490230208658, 138.23136107104716, 160.69678033606021, 182.
69000801767106, 203.3004004123143, 221.83893811890553, 237.9088727981478]
```

In [27]:

```
plt.figure(7)
plt.plot(YV,PV,'b^')
plt.plot(Year,Pop,'ro')
plt.axis([1780,2000,0,260])
plt.title('Population of US')
plt.show()
```



We can see that the Verhulst model provides a very accurate estimation of US population from 1790 until 1925. The higher estimation than the actual US population from 1925 to 1990 may be due to unexpected factors effecting the population size like the First and Second World Wars and the Great Depression.

Q.3 (*) Cohort Model

Consider the human population divided into three age groups $0-14$, $15-39$ and ≥ 40 with population size P_i , $i=1,2,3$ and yearly birth rates of 0, 0.06 and 0 and death rates of 0.005, 0.01 and 0.015, respectively.

(a). Show that $\mathbf{P}(t+1)=\mathbf{A}\mathbf{P}(t)$ where

$$\mathbf{P}=\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix}, \quad \mathbf{A}=\begin{bmatrix} 0.92867 & 0.06 & 0 \\ 0.06\,63 & 0.9504 & 0 \\ 0 & 0.0\,396 & 0.985 \end{bmatrix}.$$

(b). Find the largest eigenvalue of $\mathbf{A}$ and describe the long-term behaviour of this system.

(c). Consider the growth of the population with the following initial populations. How long does it take for the total population to double in each case?

(i) $P_1=200$, $P_2=P_3=400$.

(ii) $P_1=400$, $P_2=P_3=300$.

(iii) $P_1=200$, $P_2=500$ and $P_3=300$.

Explain what you see.

(a)

$$P_1(t+1)=b_1P_1(t)+\tfrac{14}{15}(1-d_1)P_1(t)+b_2P_2(t)+b_3P_3(t)=0.92867P_1(t)+0.06P_2($$

$$P_2(t+1)=\tfrac{1}{15}(1-d_1)P_1(t)+\tfrac{24}{25}(1-d_2)P_2(t)=0.0663P_1(t)+0.9504P_2(t)$$

$$P_3(t+1)=\tfrac{1}{25}(1-d_2)P_2(t)+(1-d_3)P_3(t)=0.0396P_2(t)+0.985P_3(t)$$

In [31]:

```
b=[0, 0.06 ,0]
d=[0.005, 0.01, 0.015]
```

In [32]:

```
A=np.mat([[b[0] + 14/15 - 14/15 * d[0], b[1], b[2]],
          [1/15*(1-d[0]), 24/25*(1-d[1]), 0],
          [0, 1/25*(1-d[1]), 1-d[2]]])
# recall Python indexing begins at 0
```

In [33]:

```
print(A)
```

```
[[0.92866667 0.06       0.         ]
 [0.06633333 0.9504     0.         ]
 [0.         0.0396     0.985      ]]
```

$P(t+1)=AP(t)$

Find eigenvalues and eigenvectors

In [34]:
```
X, V=np.linalg.eig(A)
print(X) # eigen values
print(V) # columns are right eigenvectors
```

```
[0.985      1.00354962 0.87551705]
[[ 0.          0.32180453  0.72790385]
 [ 0.          0.40162787 -0.64479684]
 [ 1.          0.85740124  0.23322312]]
```

(b) Can extract first, second, third eigenvalues/vectors as follows:

In [35]:
```
lambda0=X[0]
u0=V[:,0] # This is a 3 by 1 matrix
print(lambda0,u0)
```

```
0.985 [[0.]
 [0.]
 [1.]]
```

In [36]:
```
np.matmul(A,u0)/lambda0 # A u = lambda u
```

Out[36]:
```
matrix([[0.],
        [0.],
        [1.]])
```

In [37]:
```
lambda1=X[1]
u1=V[:,1]
print(lambda1,u1)
```

```
1.0035496159842685 [[0.32180453]
 [0.40162787]
 [0.85740124]]
```

In [38]:
```
np.matmul(A,u1)/lambda1 # A u = lambda u
```

Out[38]:
```
matrix([[0.32180453],
        [0.40162787],
        [0.85740124]])
```

In [39]:
```
lambda2=X[2]
u2=V[:,2]
print(lambda2,u2)
```

```
0.8755170506823979 [[ 0.72790385]
 [-0.64479684]
 [ 0.23322312]]
```

In [40]:
```
np.matmul(A,u2)/lambda2 # A u = lambda u
```

Out[40]:
```
matrix([[ 0.72790385],
        [-0.64479684],
        [ 0.23322312]])
```

In [41]:
```
max(lambda0, lambda1, lambda2)
```

Out[41]:
```
1.0035496159842685
```

We can see that the largest eigenvalue is lambda1 with a value of 1.0035496159842685.

As this is greater than 1, slow exponential growth will occur in this model

(c) Initial poplulation $P_1 = P_2 = P_3 = 100$.

In [42]:
```
P0=[100,100,100]
```

In [43]:
```
np.matmul(A,P0) # population after one year
```

Out[43]:
```
matrix([[ 98.86666667, 101.67333333, 102.46      ]])
```

In [44]:
```
n=20 # 20 years
An=np.linalg.matrix_power(A,n) # matrix A^n
print(An)
```

```
[[0.48657128 0.47022728 0.        ]
 [0.51986238 0.65689805 0.        ]
 [0.24436408 0.51787528 0.73913643]]
```

In [62]:
```
Pn=np.matmul(An,P0) # population after n years
print(Pn)
```

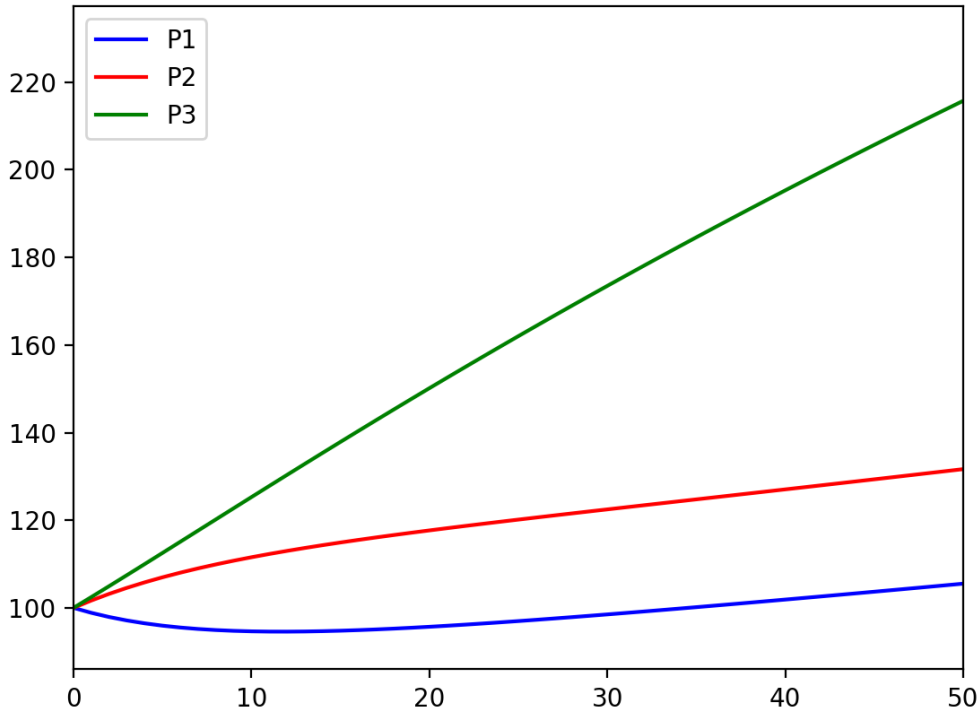
```
[[ 95.67985614 117.67604324 150.13757869]]
```

In [64]:
```
Pn.sum()
```

Out[64]:
```
363.49347806660535
```

Define lists P1, P2, P3 giving population at time 0,1,2,3, etc

In [51]:
```
nit=50 # nit is no of time steps

#initialise list to zero

P1=[0]*(nit+1)
P2=[0]*(nit+1)
P3=[0]*(nit+1)

#Given initial population

P1[0],P2[0],P3[0]=100,100,100

# compute P1,P2,P3
for n in range(1,nit+1):
    AP=np.matmul(A,[P1[n-1],P2[n-1],P3[n-1]]) # AP[n-1]
    P1[n],P2[n],P3[n]=AP[0,0],AP[0,1],AP[0,2]

# plotting

t=np.arange(0,nit+1,1) #  time steps
plt.figure()
P1plot,=plt.plot(t,P1,'b-')
P2plot,=plt.plot(t,P2,'r-')
P3plot,=plt.plot(t,P3,'g-')
plt.legend((P1plot,P2plot,P3plot),('P1','P2','P3'))
Pmax=max(max(P1),max(P2),max(P3))
Pmin=min(min(P1),min(P1),min(P1))
plt.axis([0,nit,Pmin/1.1,Pmax*1.1])
plt.show()
```



Compare populations to eigenvector for largest eigenvalue

In [52]:
```
print(P2[50]/P1[50],u1[1,0]/u1[0,0]) # compare first and second cohorts to
```

```
1.247744063179402 1.248049155293366
```

In [53]:
```
print(P3[50]/P1[50],u1[2,0]/u1[0,0]) # compare first and third cohorts to e
```

```
2.0444670328229053 2.664354161915348
```

(c)

(i) $P_1 = 200, P_2 = P_3 = 400$

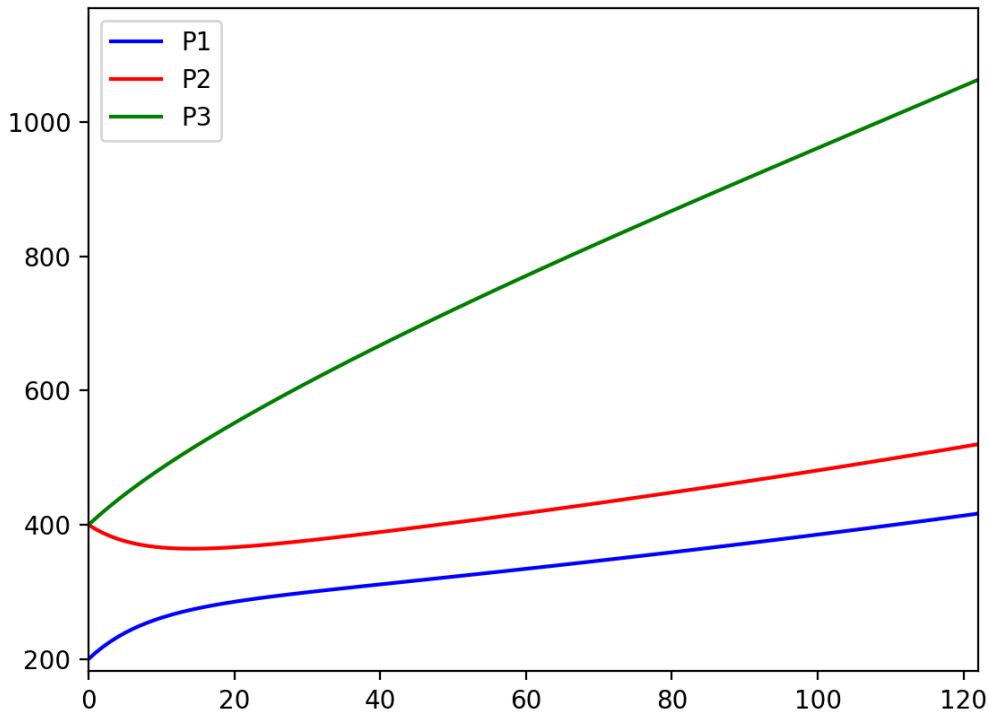
In [54]:
```
P0=[200,400,400]

totalPopulation = sum(P0)
```

In [55]:
```
years = 0
for n in range(300):
    An=np.linalg.matrix_power(A,n) # matrix A^n
    Pn=np.matmul(An,P0) # population after n years
    if Pn.sum() > totalPopulation*2:
        years = n
        break;
print("The total population doubles after ", n, " years.")
```

```
The total population doubles after  122  years.
```

In [56]:

```
nit=122 # nit is no of time steps

#initialise list to zero

P1=[0]*(nit+1)
P2=[0]*(nit+1)
P3=[0]*(nit+1)

#Given initial population

P1[0],P2[0],P3[0]=P0[0], P0[1], P0[2]

# compute P1,P2,P3
for n in range(1,nit+1):
    AP=np.matmul(A,[P1[n-1],P2[n-1],P3[n-1]]) # AP[n-1]
    P1[n],P2[n],P3[n]=AP[0,0],AP[0,1],AP[0,2]

# plotting

t=np.arange(0,nit+1,1) #  time steps
plt.figure()
P1plot,=plt.plot(t,P1,'b-')
P2plot,=plt.plot(t,P2,'r-')
P3plot,=plt.plot(t,P3,'g-')
plt.legend((P1plot,P2plot,P3plot),('P1','P2','P3'))
Pmax=max(max(P1),max(P2),max(P3))
Pmin=min(min(P1),min(P1),min(P1))
plt.axis([0,nit,Pmin/1.1,Pmax*1.1])
plt.show()
```



In [57]:

```
print(P2[122]/P1[122],u1[1,0]/u1[0,0]) # compare first and second cohorts t
```

```
1.248049187924413 1.248049155293366
```

In [58]:

```
print(P3[122]/P1[122],u1[2,0]/u1[0,0]) # compare first and third cohorts to
```

```
2.5512020932531168 2.664354161915348
```

(c)

(ii) $P_1 = 400, P_2 = P_3 = 300$

In [62]:

```
P0=[400,300,300]

totalPopulation = sum(P0)
print("Starting population: ",totalPopulation)
```

```
Starting population:  1000
```

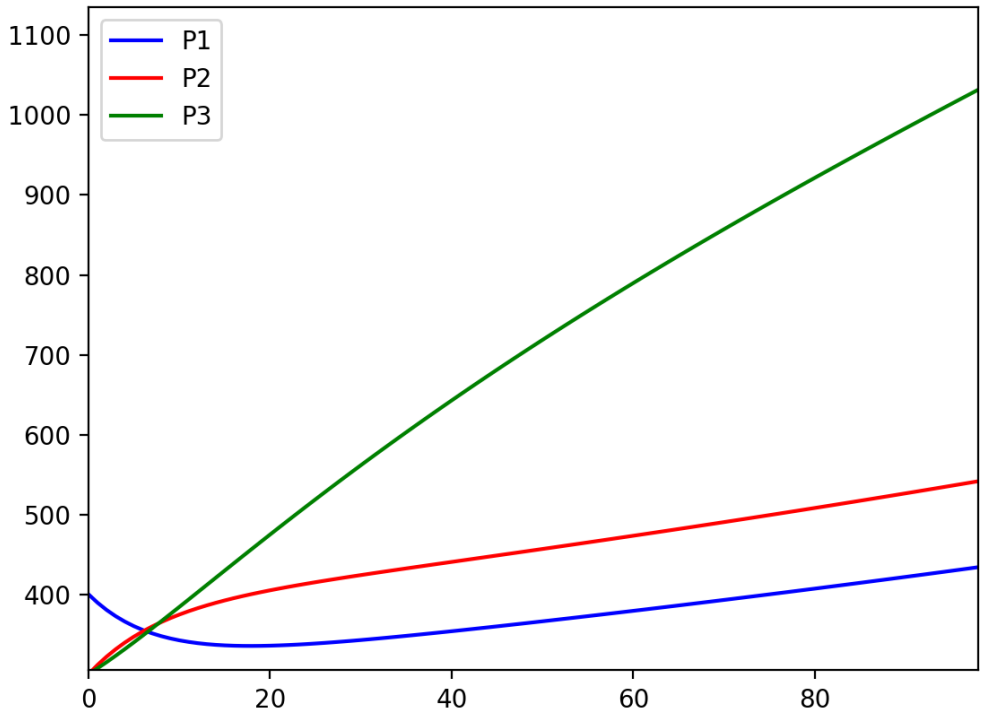
In [63]:

```
years = 0
for n in range(300):
    An=np.linalg.matrix_power(A,n) # matrix A^n
    Pn=np.matmul(An,P0) # population after n years
    if Pn.sum() > totalPopulation*2:
        years = n
        break;
print("The total population doubles after ", n, " years.")
```

```
The total population doubles after  98  years.
```

In [64]:

```
nit=98 # nit is no of time steps

#initialise list to zero

P1=[0]*(nit+1)
P2=[0]*(nit+1)
P3=[0]*(nit+1)

#Given initial population

P1[0],P2[0],P3[0]=P0[0], P0[1], P0[2]

# compute P1,P2,P3
for n in range(1,nit+1):
    AP=np.matmul(A,[P1[n-1],P2[n-1],P3[n-1]]) # AP[n-1]
    P1[n],P2[n],P3[n]=AP[0,0],AP[0,1],AP[0,2]

# plotting

t=np.arange(0,nit+1,1) #  time steps
plt.figure()
P1plot,=plt.plot(t,P1,'b-')
P2plot,=plt.plot(t,P2,'r-')
P3plot,=plt.plot(t,P3,'g-')
plt.legend((P1plot,P2plot,P3plot),('P1','P2','P3'))
Pmax=max(max(P1),max(P2),max(P3))
Pmin=min(min(P1),min(P1),min(P1))
plt.axis([0,nit,Pmin/1.1,Pmax*1.1])
plt.show()
```



In [66]:
```
print(P2[98]/P1[98],u1[1,0]/u1[0,0]) # compare first and second cohorts to
```

```
1.248048146439516 1.248049155293366
```

In [67]:
```
print(P3[98]/P1[98],u1[2,0]/u1[0,0]) # compare first and third cohorts to e
```

```
2.377783277474929 2.664354161915348
```

(iii) $P_1 = 200, P_2 = 500, P_3 = 300$

In [69]:
```
P0=[200,500,300]

totalPopulation = sum(P0)
print("Starting population: ",totalPopulation)
```

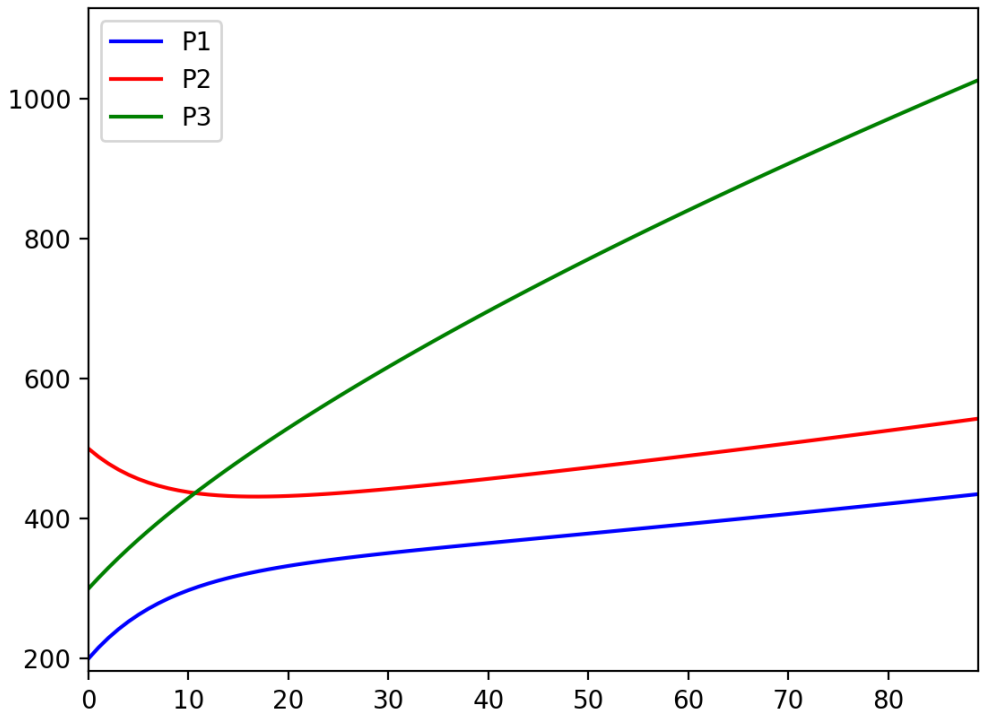
```
Starting population:  1000
```

In [70]:
```
years = 0
for n in range(300):
    An=np.linalg.matrix_power(A,n) # matrix A^n
    Pn=np.matmul(An,P0) # population after n years
    if Pn.sum() > totalPopulation*2:
        years = n
        break;
print("The total population doubles after ", n, " years.")
```

```
The total population doubles after  89  years.
```

In [71]:

```
nit=89 # nit is no of time steps

#initialise list to zero

P1=[0]*(nit+1)
P2=[0]*(nit+1)
P3=[0]*(nit+1)

#Given initial population

P1[0],P2[0],P3[0]=P0[0], P0[1], P0[2]

# compute P1,P2,P3
for n in range(1,nit+1):
    AP=np.matmul(A,[P1[n-1],P2[n-1],P3[n-1]]) # AP[n-1]
    P1[n],P2[n],P3[n]=AP[0,0],AP[0,1],AP[0,2]

# plotting

t=np.arange(0,nit+1,1) #  time steps
plt.figure()
P1plot,=plt.plot(t,P1,'b-')
P2plot,=plt.plot(t,P2,'r-')
P3plot,=plt.plot(t,P3,'g-')
plt.legend((P1plot,P2plot,P3plot),('P1','P2','P3'))
Pmax=max(max(P1),max(P2),max(P3))
Pmin=min(min(P1),min(P1),min(P1))
plt.axis([0,nit,Pmin/1.1,Pmax*1.1])
plt.show()
```



In [73]:

```
print(P2[89]/P1[89],u1[1,0]/u1[0,0]) # compare first and second cohorts to
```

```
1.248053340193975 1.248049155293366
```

In [74]:

```
print(P3[89]/P1[89],u1[2,0]/u1[0,0]) # compare first and third cohorts to e
```

```
2.3601791730634036 2.664354161915348
```

We can see from the three different initial populations that regardless of initial population, over time P1, P2 and P3 grow exponentially, in line with the dominant eigenvalue of 1.004

Q.4 (*) Chaotic Models

Consider the Ricker model of Salmon population size as follows

$$P_{n+1} = re^{-\frac{P_n}{K}} P_n$$

where r is the rate and K is another parameter. For simplicity choose $K = 500$ and consider the behaviour of this system for

(a). $r < 1$, with initial population of 1000.

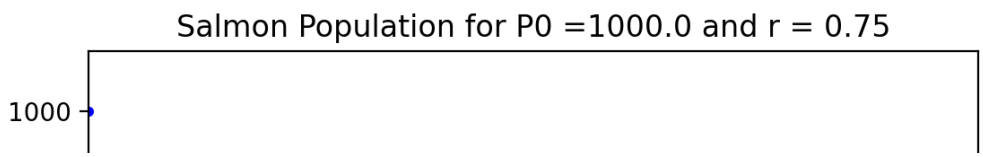
(b). $1 < r < e^2$, for initial population 100, 500 and 1000.

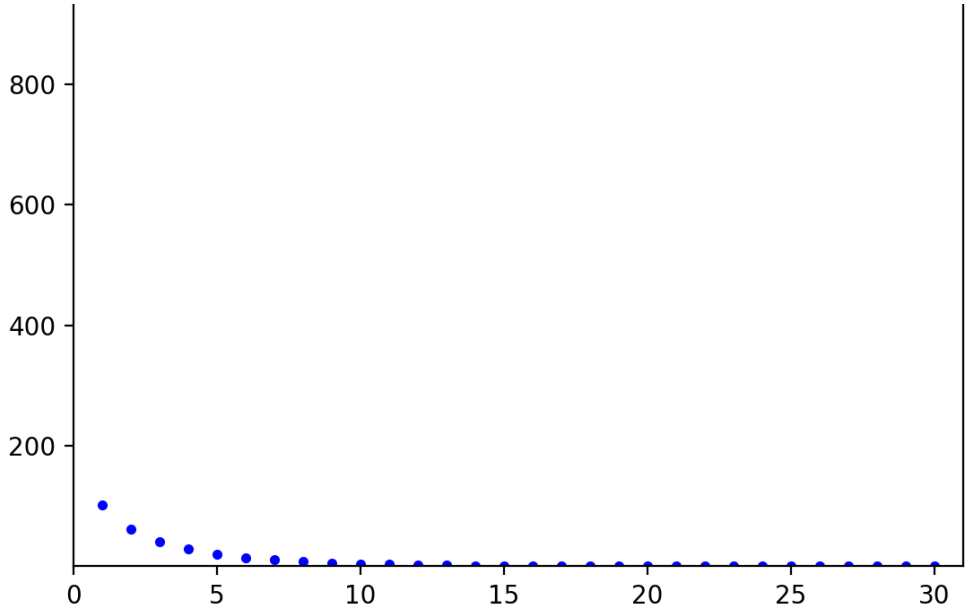
(c). $r > e^2$, for initial population 100, 500 and 1000.

Explain what you see.

In [75]:

```
# initial values
P_0=1000.
K=500.
r=0.75

nit=30 # nit is no of time steps

t=np.arange(0,nit+1,1) #  time steps

P=[0]*(nit+1) # list for P initialized with nit zeros

P[0]=P_0 # given initial population
for n in range(1,nit+1):
    P[n]=r*P[n-1]*np.exp(-P[n-1]/K)

# plotting
plt.figure()
plt.plot(t,P,'b.')
Pmax=max(P)
Pmin=min(P)
plt.axis([0,nit+1,Pmin/1.1,Pmax*1.1])
plt.title('Salmon Population for P0 =' +str(P_0)+ ' and r = ' +str(r))
plt.show()
```





Population dies out for $r < 1$.

Explore the other values of r where one can find convergence to non-zero population for $1 < r < e^2$ and period doubling phenomena and chaos for $r > e^2$.

In [76]:

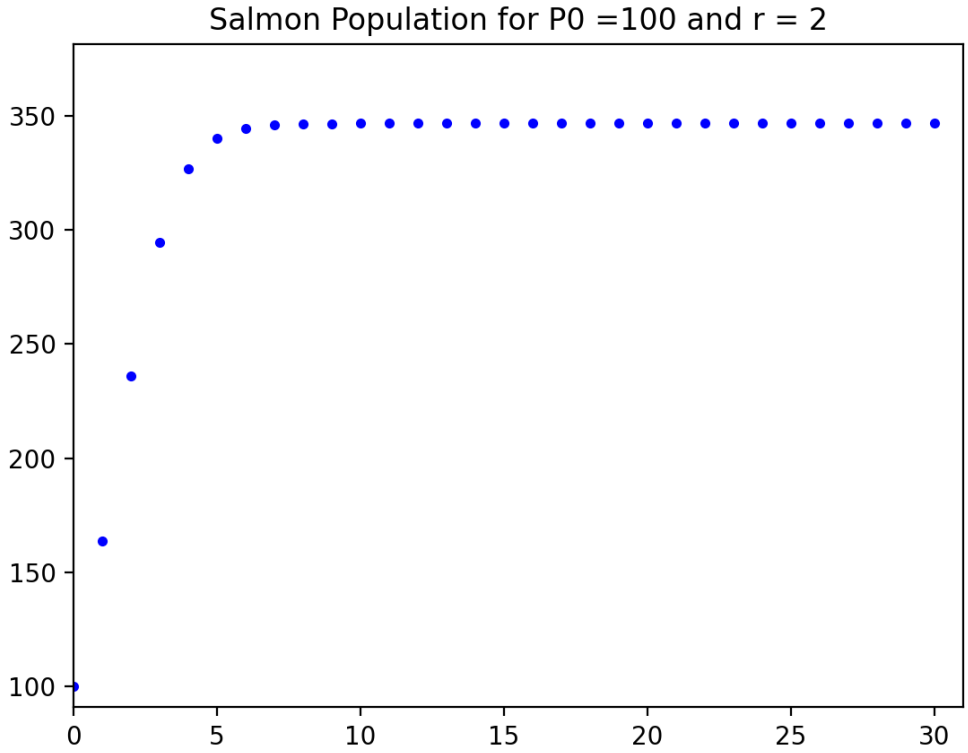
```
e2=np.exp(2)
print(e2)
```

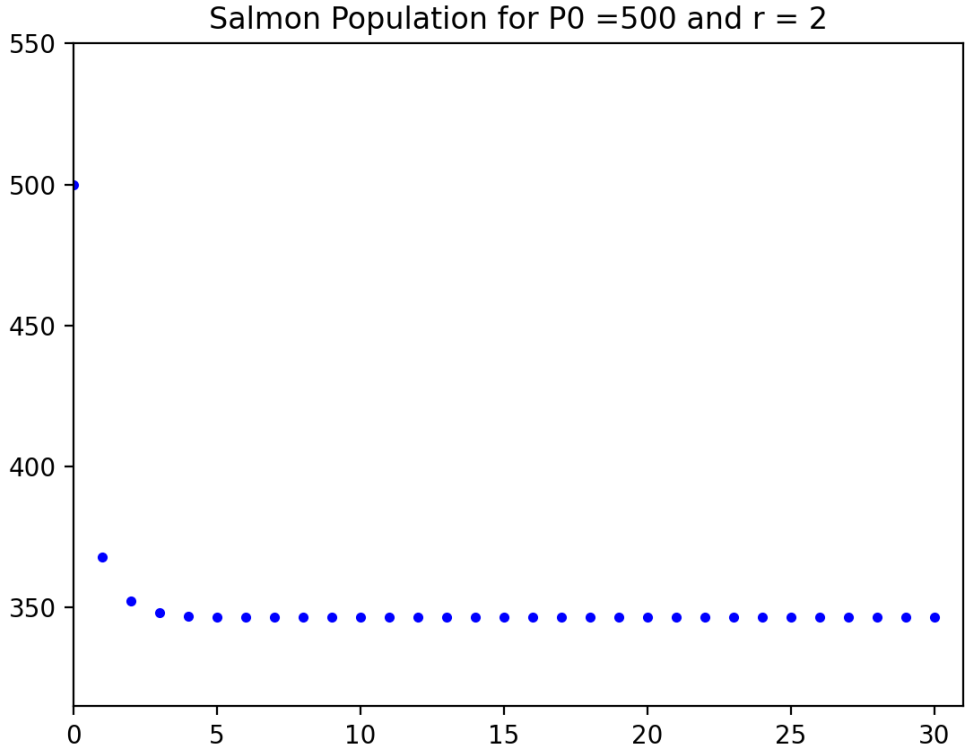
```
7.38905609893065
```

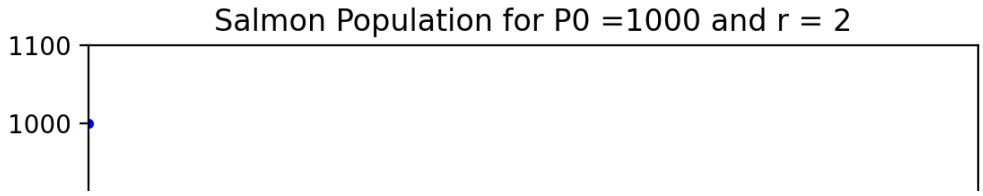
(b). $1 < r < e^2$, for initial population 100, 500 and 1000.

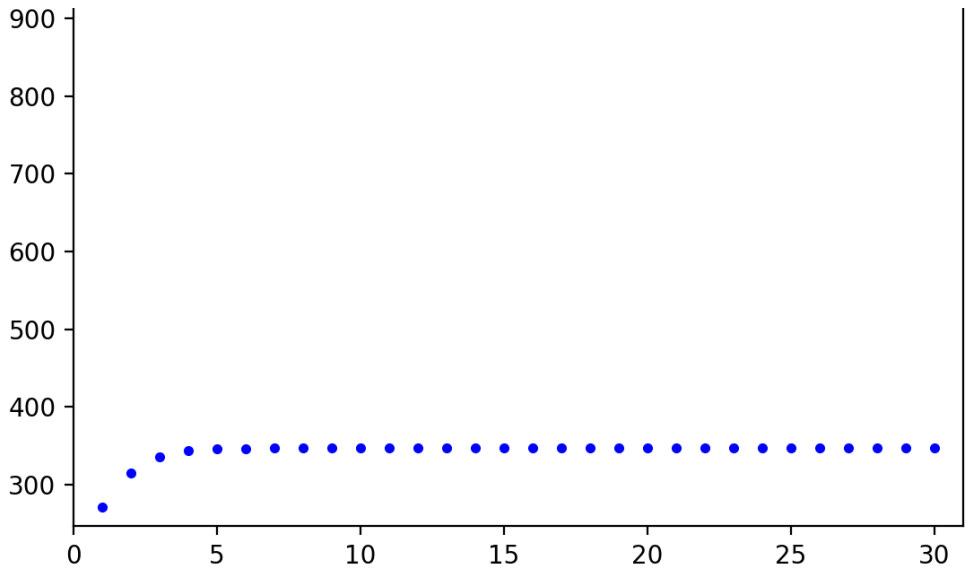
In [103…

```
#r = 2
populations = [100, 500, 1000]
for i in range(3):
    # initial values
    P_0=populations[i]
    K=500
    r=2

    nit=30 # nit is no of time steps

    t=np.arange(0,nit+1,1) #  time steps

    P=[0]*(nit+1) # list for P initialized with nit zeros

    P[0]=P_0 # given initial population
    for n in range(1,nit+1):
        P[n]=r*P[n-1]*np.exp(-P[n-1]/K)

    # plotting
    plt.figure()
    plt.plot(t,P,'b.')
    Pmax=max(P)
    Pmin=min(P)
    plt.axis([0,nit+1,Pmin/1.1,Pmax*1.1])
    plt.title('Salmon Population for P0 =' +str(P_0)+ ' and r = ' +str(r))
    plt.show()
```


Salmon Population for P0 =100 and r = 2
350
300
250
200
150
100
0
5
10
15
20
25
30


Salmon Population for P0 =500 and r = 2
550
500
450
400
350
0
5
10
15
20
25
30

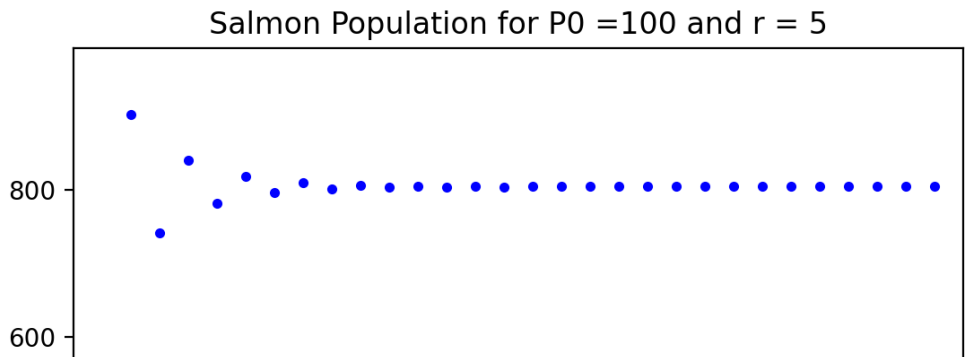

Salmon Population for P0 =1000 and r = 2
1100
1000

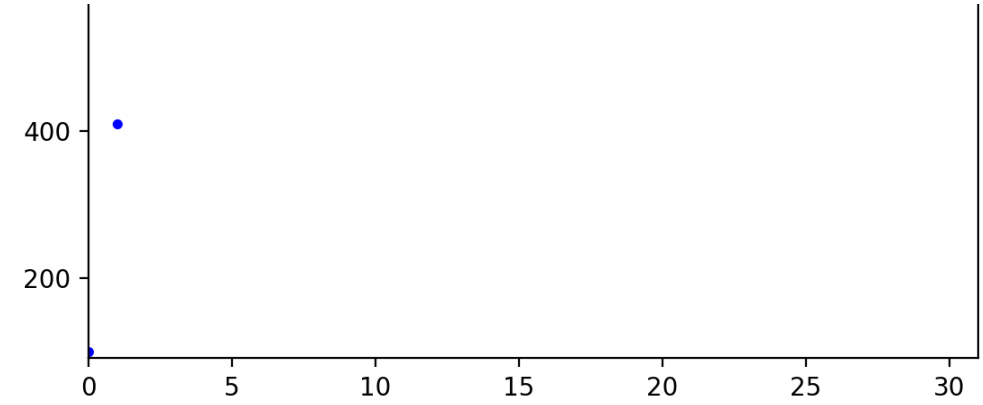


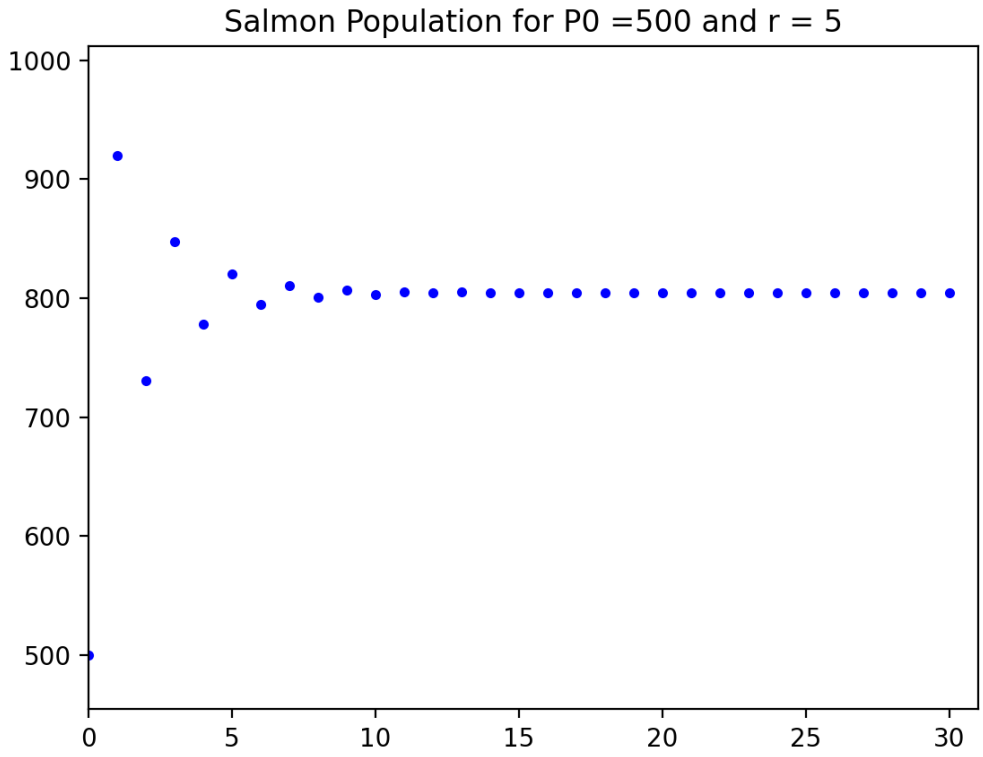
When R_0 = 2, the population converges to $P^* = K \ln R_0 = (500)(\ln 2) = 346.57$

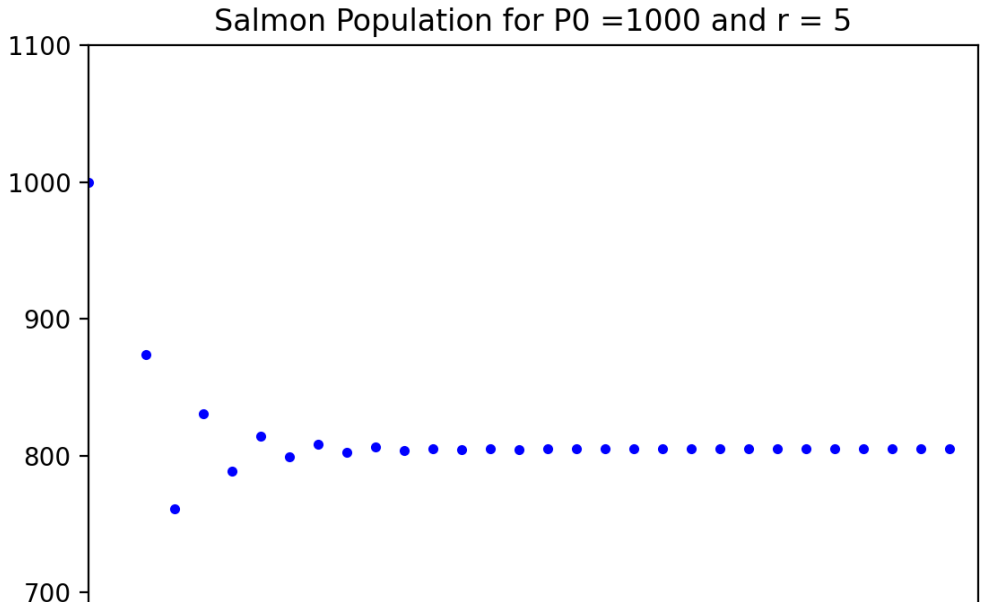
In [108…

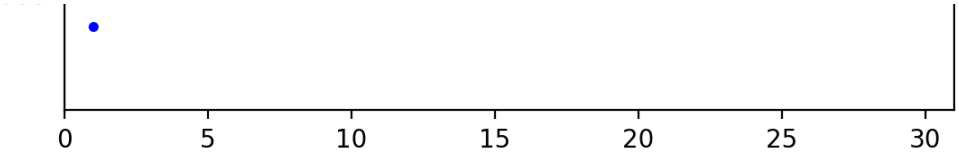
```
#r = 5
populations = [100, 500, 1000]
for i in range(3):
    # initial values
    P_0=populations[i]
    K=500
    r=5

    nit=30 # nit is no of time steps

    t=np.arange(0,nit+1,1) #  time steps

    P=[0]*(nit+1) # list for P initialized with nit zeros

    P[0]=P_0 # given initial population
    for n in range(1,nit+1):
        P[n]=r*P[n-1]*np.exp(-P[n-1]/K)

    # plotting
    plt.figure()
    plt.plot(t,P,'b.')
    Pmax=max(P)
    Pmin=min(P)
    plt.axis([0,nit+1,Pmin/1.1,Pmax*1.1])
    plt.title('Salmon Population for P0 =' +str(P_0)+ ' and r = ' +str(r))
    plt.show()
```




400
200
0
5
10
15
20
25
30


Salmon Population for P0 =500 and r = 5
1000
900
800
700
600
500
0
5
10
15
20
25
30


Salmon Population for P0 =1000 and r = 5
1100
1000
900
800
700

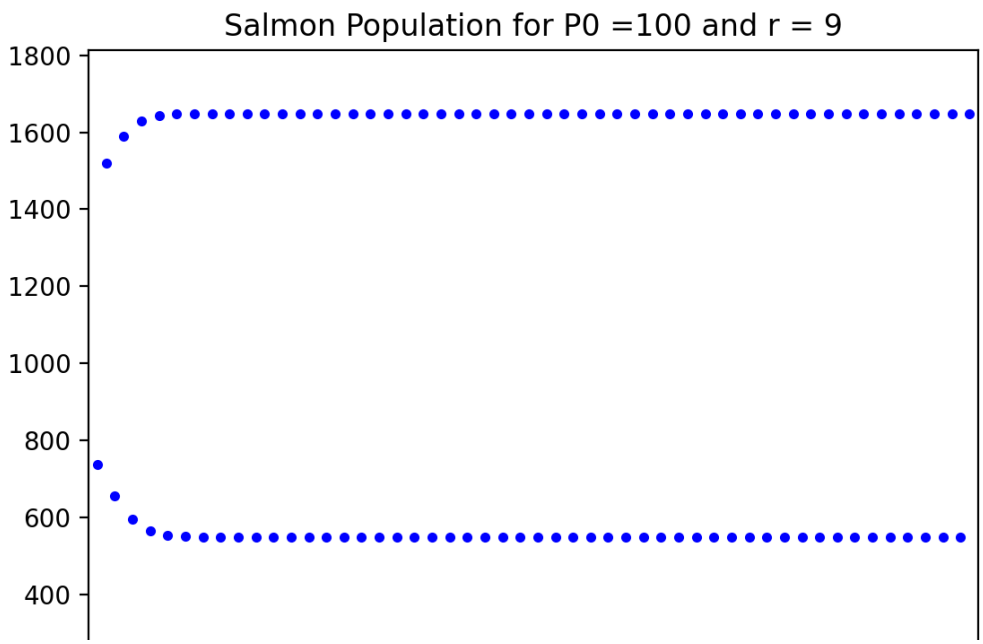


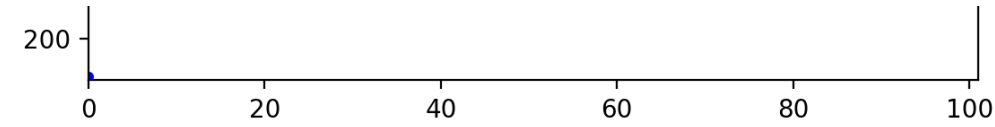
When R_0 = 5, the population converges to $P^* = K \ln R_0 = (500)(\ln 5) = 804.72$

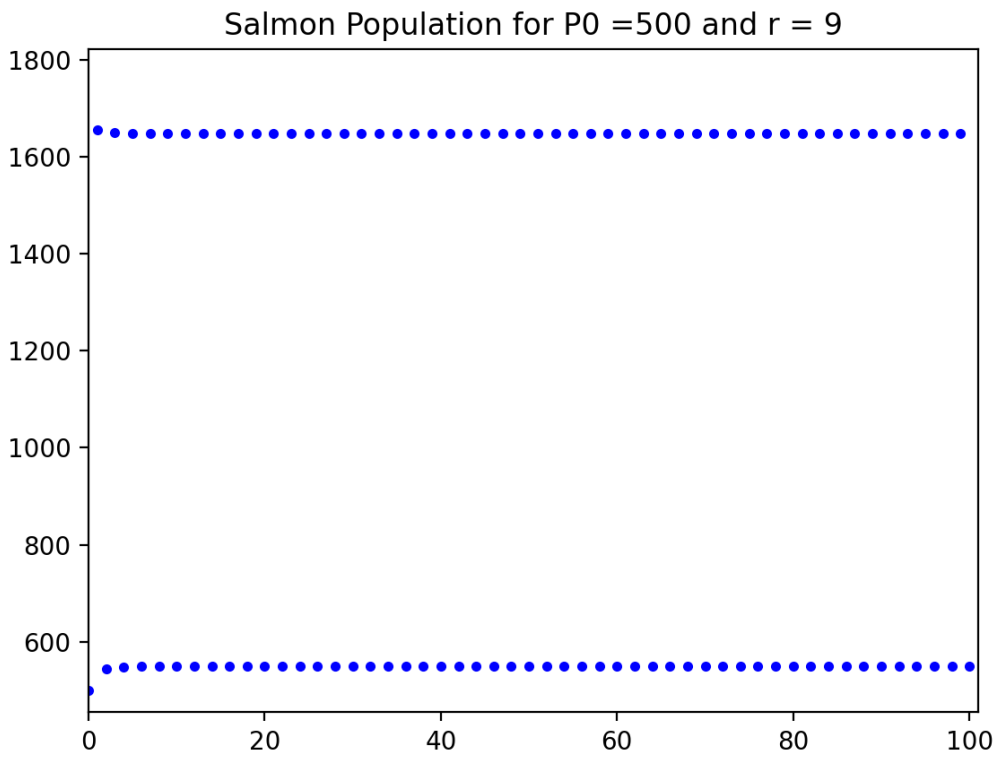
(c). $r > e^2$, for initial population 100, 500 and 1000.

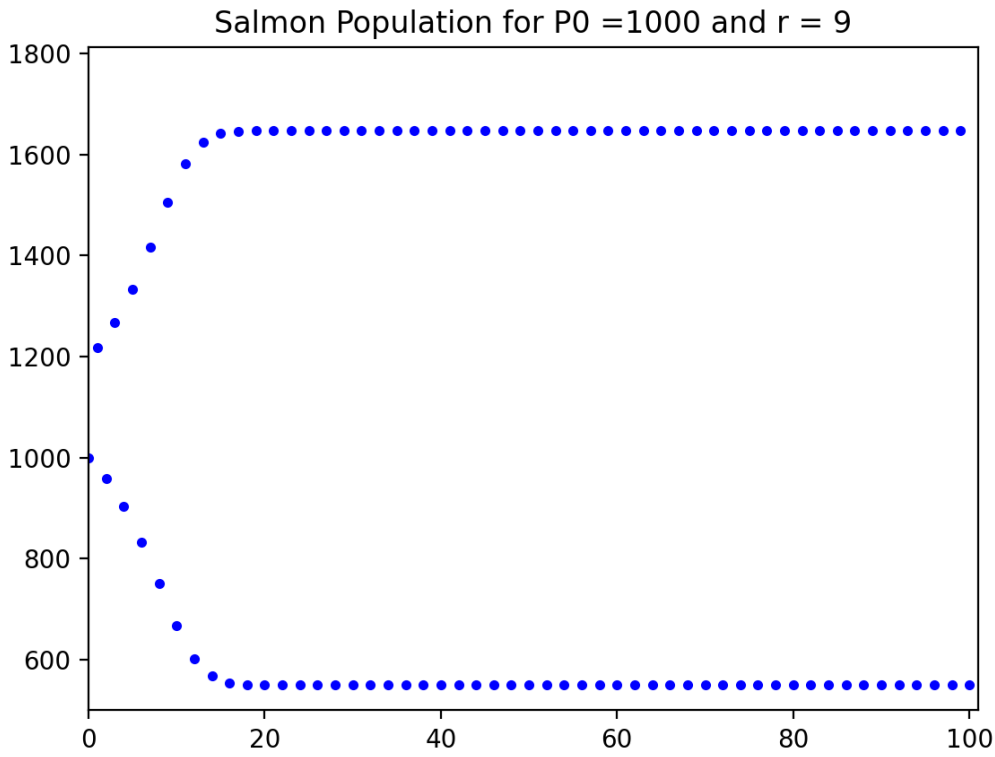
In [112…

```
#r = 9
populations = [100, 500, 1000]
for i in range(3):
    # initial values
    P_0=populations[i]
    K=500
    r=9


    nit=100 # nit is no of time steps

    t=np.arange(0,nit+1,1) #  time steps

    P=[0]*(nit+1) # list for P initialized with nit zeros

    P[0]=P_0 # given initial population
    for n in range(1,nit+1):
        P[n]=r*P[n-1]*np.exp(-P[n-1]/K)


    # plotting
    plt.figure()
    plt.plot(t,P,'b.')
    Pmax=max(P)
    Pmin=min(P)
    plt.axis([0,nit+1,Pmin/1.1,Pmax*1.1])
    plt.title('Salmon Population for P0 =' +str(P_0)+ ' and r = ' +str(r))
    plt.show()
```







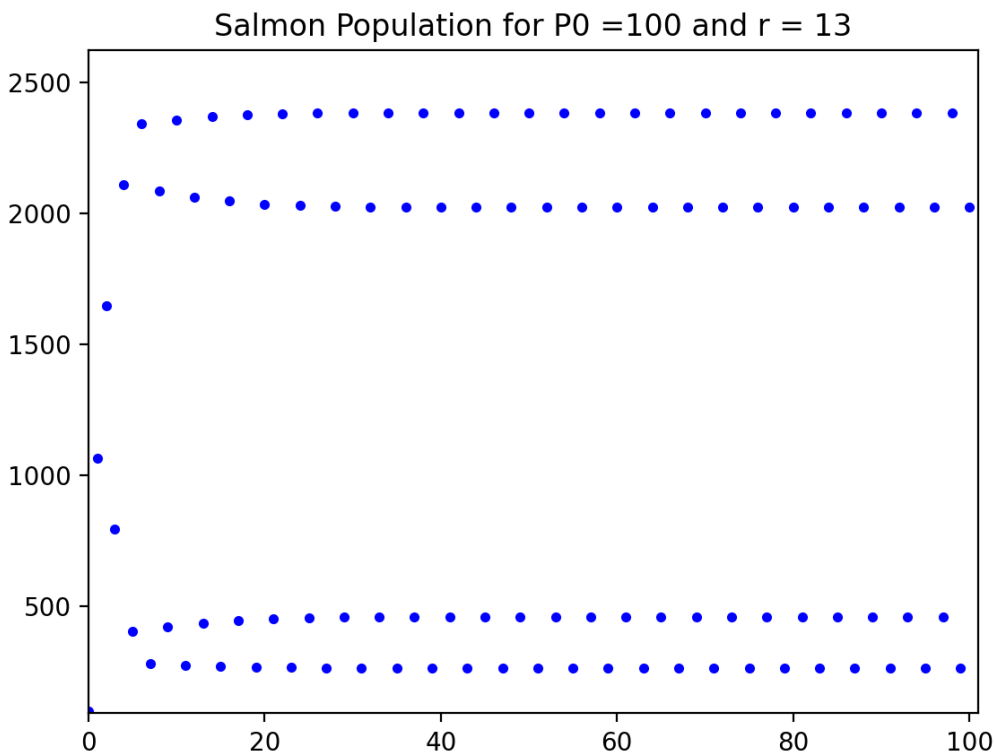


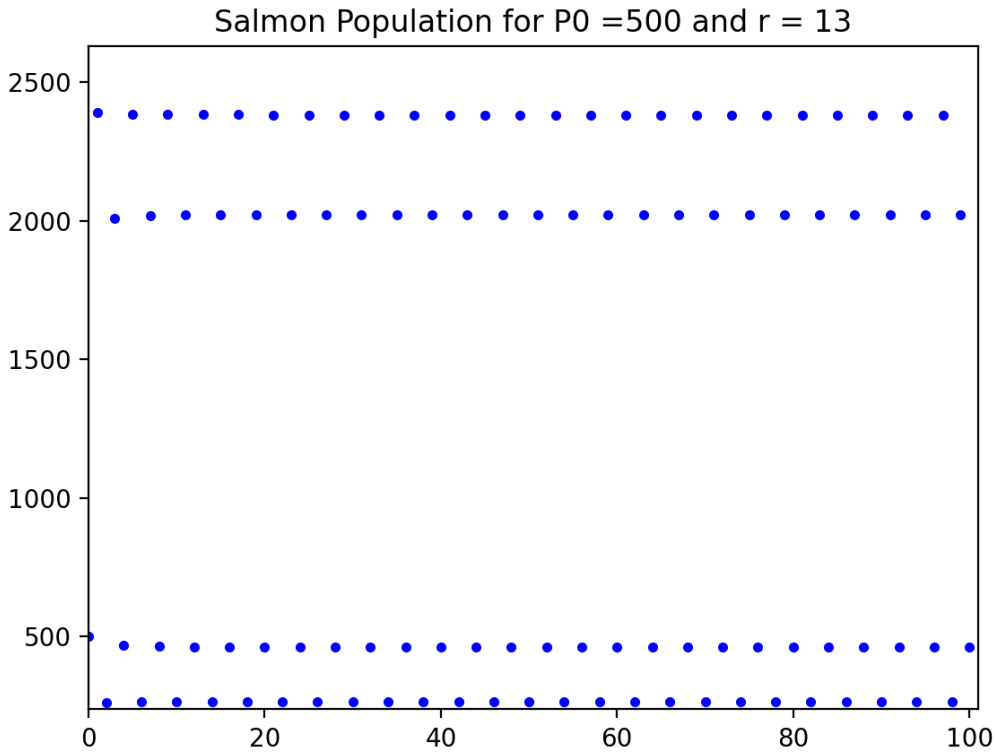
When R_0 = 9 we see that period doubling occurs with $1647.92 \approx F(549.31)$ and

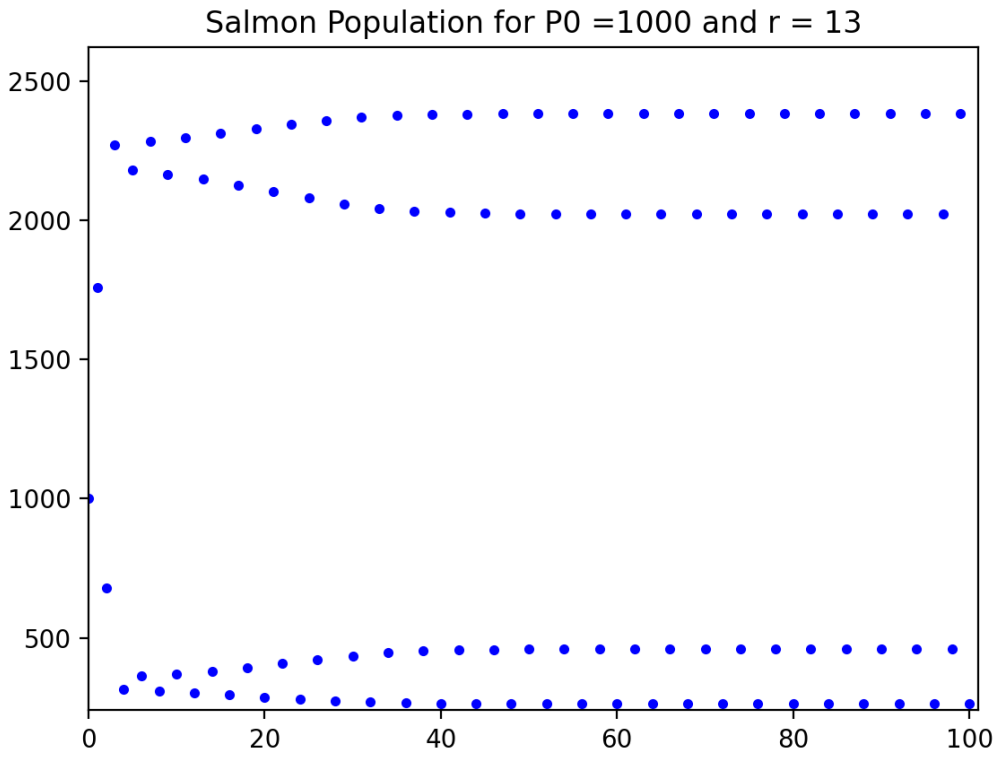
$549.31 \approx F(1647.92)$ where the population stabilises at either 1648 or 549 depending on the previous year.

In [116…

```
#r = 13
populations = [100, 500, 1000]
for i in range(3):
    # initial values
    P_0=populations[i]
    K=500
    r=13


    nit=100 # nit is no of time steps

    t=np.arange(0,nit+1,1) #  time steps

    P=[0]*(nit+1) # list for P initialized with nit zeros

    P[0]=P_0 # given initial population
    for n in range(1,nit+1):
        P[n]=r*P[n-1]*np.exp(-P[n-1]/K)


    # plotting
    plt.figure()
    plt.plot(t,P,'b.')
    Pmax=max(P)
    Pmin=min(P)
    plt.axis([0,nit+1,Pmin/1.1,Pmax*1.1])
    plt.title('Salmon Population for P0 =' +str(P_0)+ ' and r = ' +str(r))
    plt.show()
```



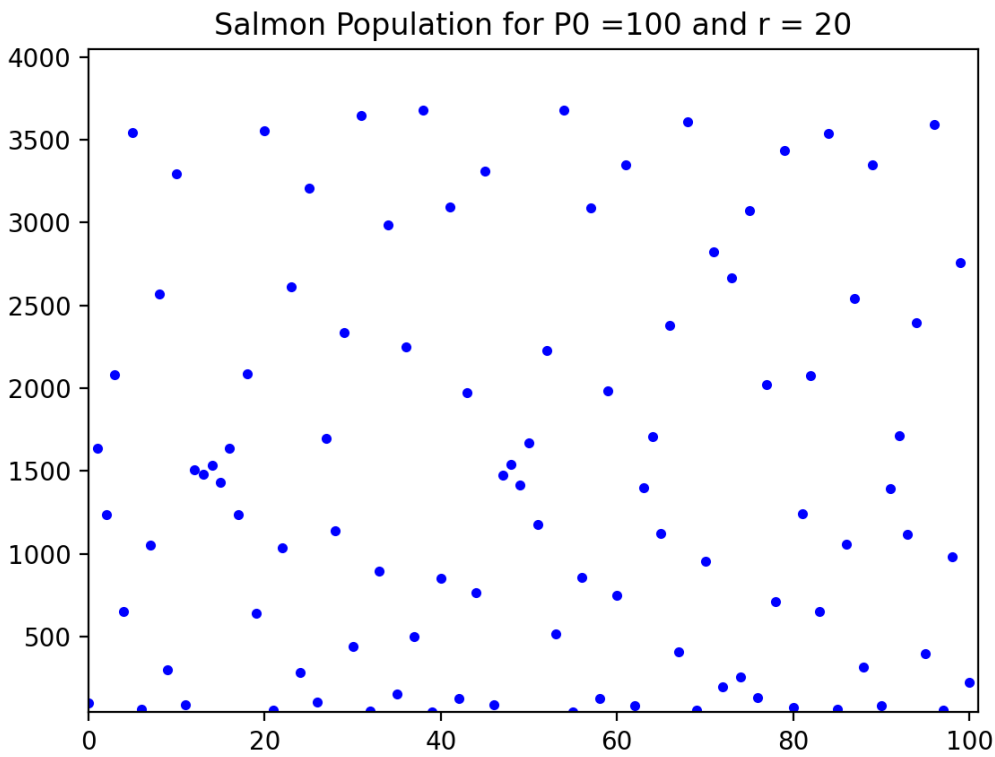


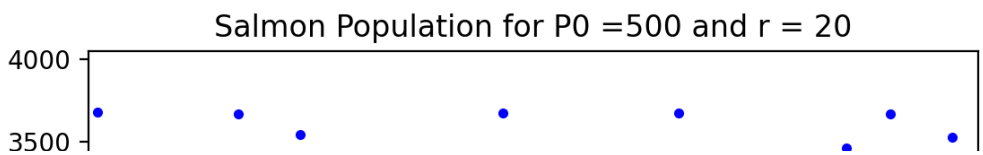


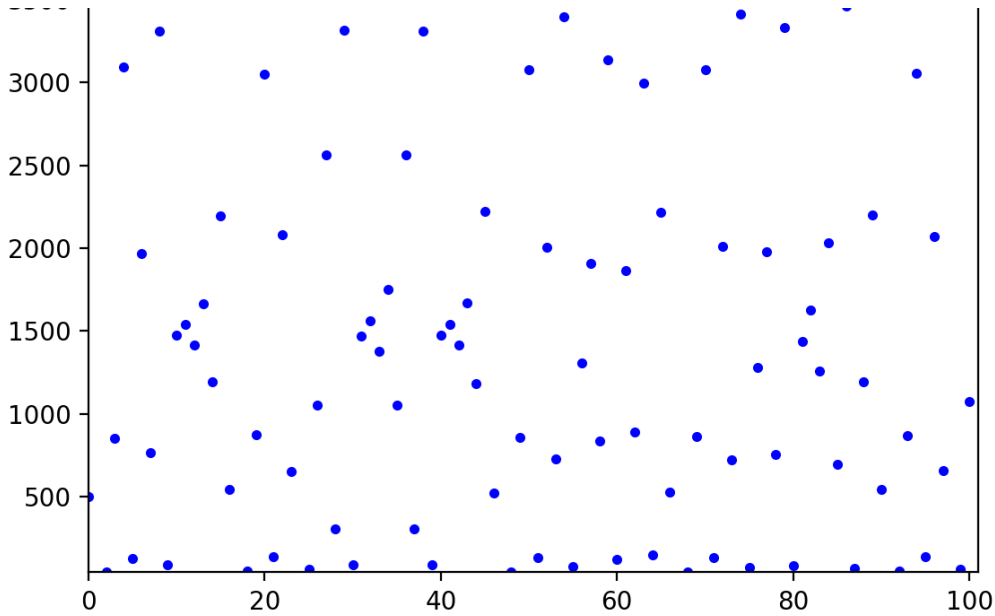
When R_0 = 13 we see further period doubling occuring.

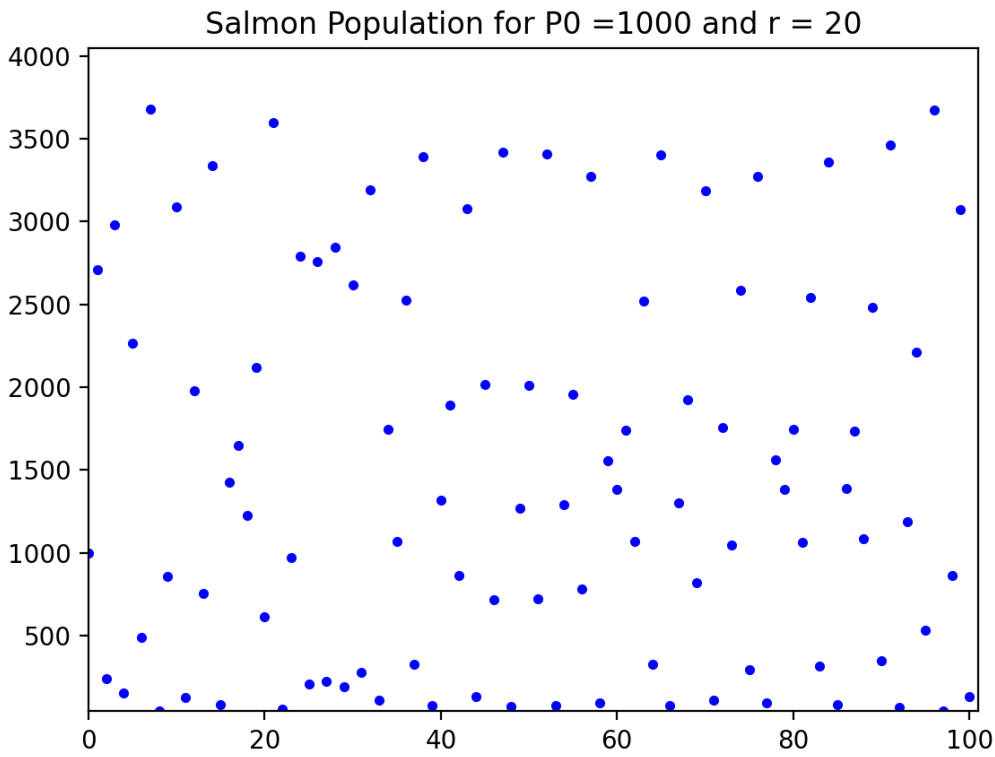
In [102…

```
#r = 20
populations = [100, 500, 1000]
for i in range(3):
    # initial values
    P_0=populations[i]
    K=500
    r=20

    nit=100 # nit is no of time steps

    t=np.arange(0,nit+1,1) #  time steps

    P=[0]*(nit+1) # list for P initialized with nit zeros

    P[0]=P_0 # given initial population
    for n in range(1,nit+1):
        P[n]=r*P[n-1]*np.exp(-P[n-1]/K)

    # plotting
    plt.figure()
    plt.plot(t,P,'b.')
    Pmax=max(P)
    Pmin=min(P)
    plt.axis([0,nit+1,Pmin/1.1,Pmax*1.1])
    plt.title('Salmon Population for P0 =' +str(P_0)+ ' and r = ' +str(r))
    plt.show()
```









When R_0 = 20 we see unpredictable chaotic behaviour.